

Modeling nonlinear relationships

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Polynomial models

Log transformations

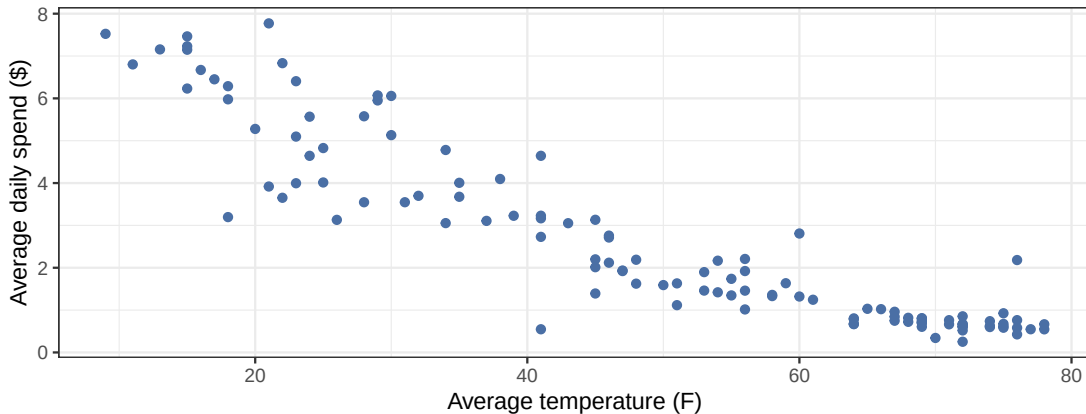
Log-log models and price elasticity



The data set `utilities` contains information on the utility bills for a house in Minnesota. We'll focus on two variables:

- `dailyspend` is the average amount of money spent on utilities (e.g. heating) for each day during the month
- `temp` is the average temperature outside for that month

What problems do you see here?



```
model <- lm(dailyspend ~ temp, data=utilities)
summary(model)
```

Call:

```
lm(formula = dailyspend ~ temp, data = utilities)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.847	-0.504	-0.024	0.515	2.448

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.34762	0.20645	35.6	<2e-16 ***
temp	-0.09643	0.00391	-24.7	<2e-16 ***

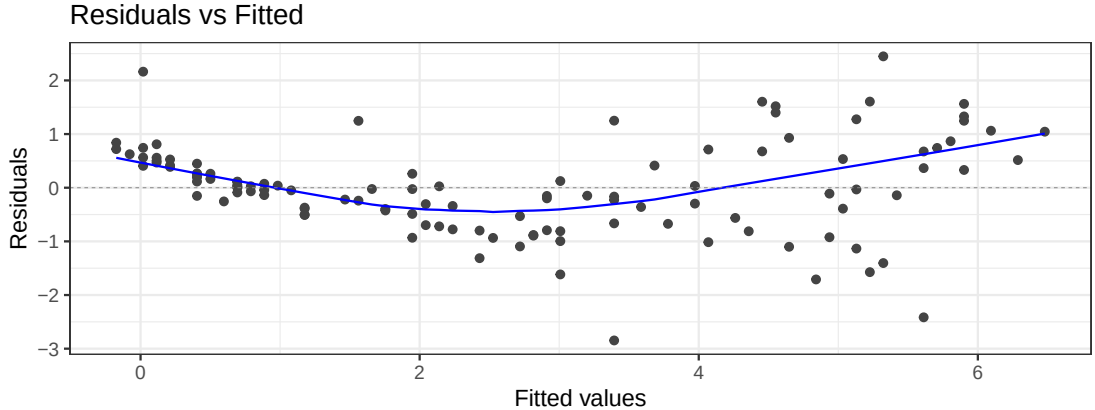
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.866 on 115 degrees of freedom

Multiple R-squared: 0.841, Adjusted R-squared: 0.84

F-statistic: 608 on 1 and 115 DF, p-value: <2e-16

Linearity and homoscedasticity are violated





- If a polynomial curve (e.g., quadratic, cubic, etc) would be a better fit for the data than a line, we can fit a curve to the data
- The way we do this is by adding X^2 , X^3 , etc. terms to the model as additional predictors

A quadratic model is still a **linear** regression — linear in the coefficients, curved in X :

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

```
poly.model <- lm(dailyspend ~ temp + I(temp^2),
                 data=utilities)
summary(poly.model)
```

Call:

```
lm(formula = dailyspend ~ temp + I(temp^2), data = utilities)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.8725	-0.2805	-0.0393	0.2639	2.1912

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.472289	0.390789	24.24	< 2e-16 ***
temp	-0.211555	0.019105	-11.07	< 2e-16 ***
I(temp^2)	0.001248	0.000204	6.12	1.3e-08 ***

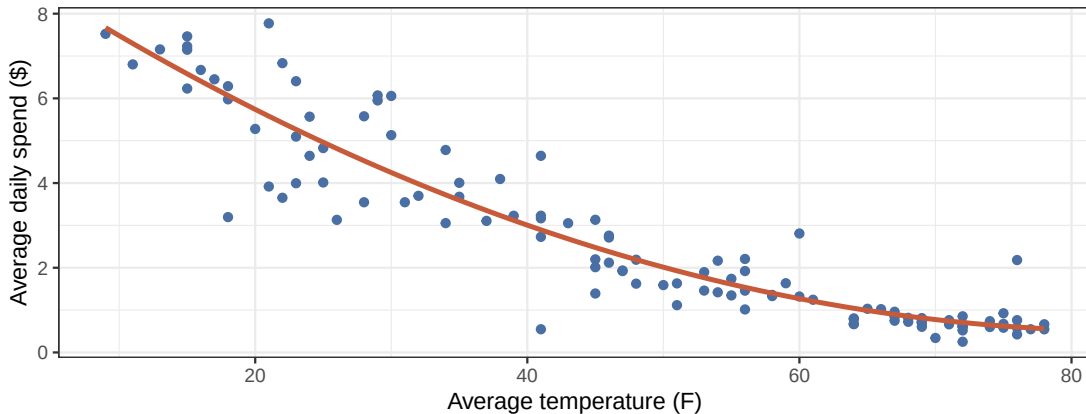
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.755 on 114 degrees of freedom

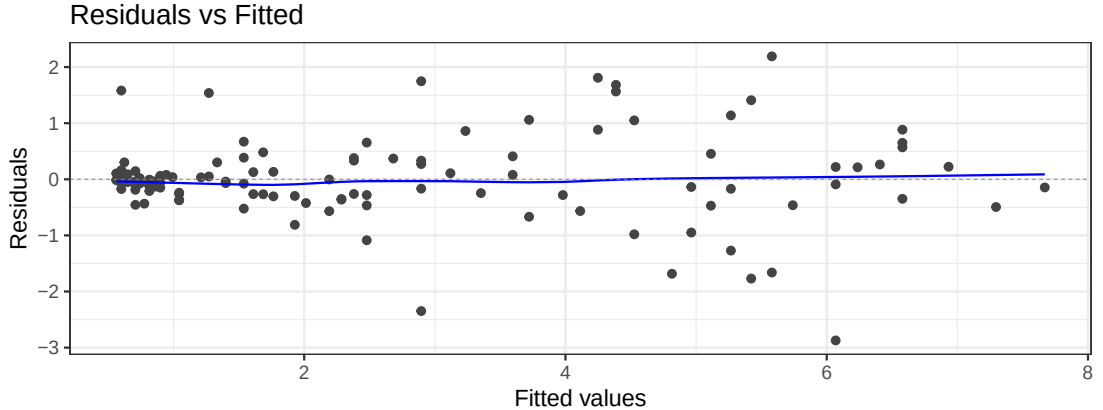
Multiple R-squared: 0.88, Adjusted R-squared: 0.878

F-statistic: 419 on 2 and 114 DF, p-value: <2e-16

Adding an X^2 term fits a parabola to the data



It solves the linearity problem!



What about a higher-order polynomial?



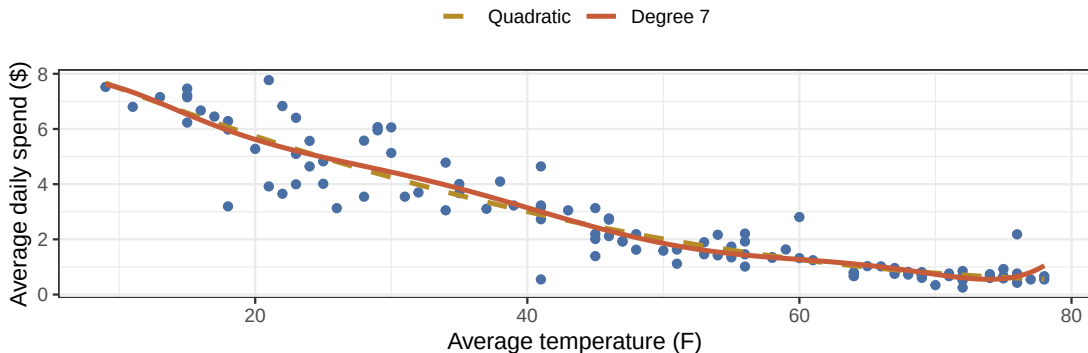
- We could fit a cubic curve by adding an X^3 term (in addition to the X and X^2 terms)
- Making the polynomial higher order will increase R^2 —why not go nuts and fit a 7th degree polynomial?

Degree		R^2	gain
1	linear	0.8410	—
2	quadratic	0.8803	+0.0394
3	cubic	0.8812	+0.0009
4	quartic	0.8813	+0.0001
5	quintic	0.8815	+0.0002
6	sextic	0.8822	+0.0006
7	septic	0.8837	+0.0015

Seven degrees buys you 0.003 of R^2



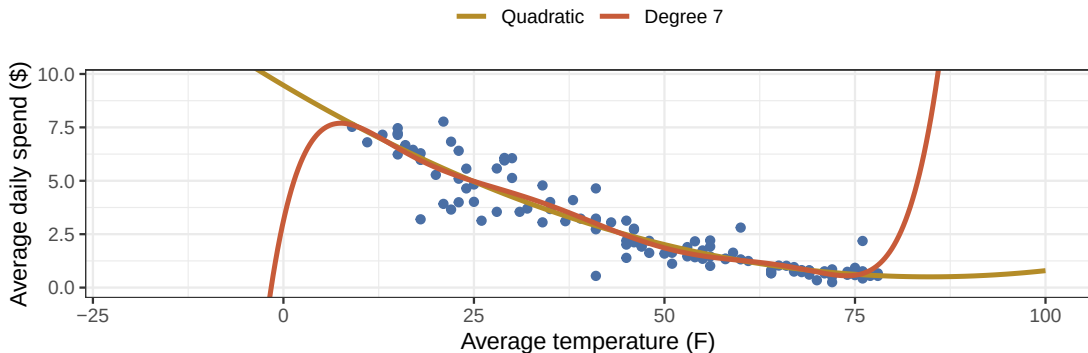
Five extra parameters, and **not one of them is significant**:



Too high a degree creates dangers with extrapolation



Both curves fit the data. Watch what happens **outside** its range:





- Start simple: only add higher-degree terms to the extent it gives you a substantial increase in R^2 , or makes an assumption hold that wasn't holding before



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- Be careful about overfitting
- Be particularly careful about extrapolating beyond the range of the data
- Mind-bender: We can think about an X^2 term as an interaction of X with itself: in a parabola, the slope depends on the value of X !

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- Sometimes, a violation of regression assumptions can be fixed by transforming one or the other of the variables (or both)
- When we transform a variable, we have to also transform our interpretation of the model
- Often the new interpretation is more meaningful!





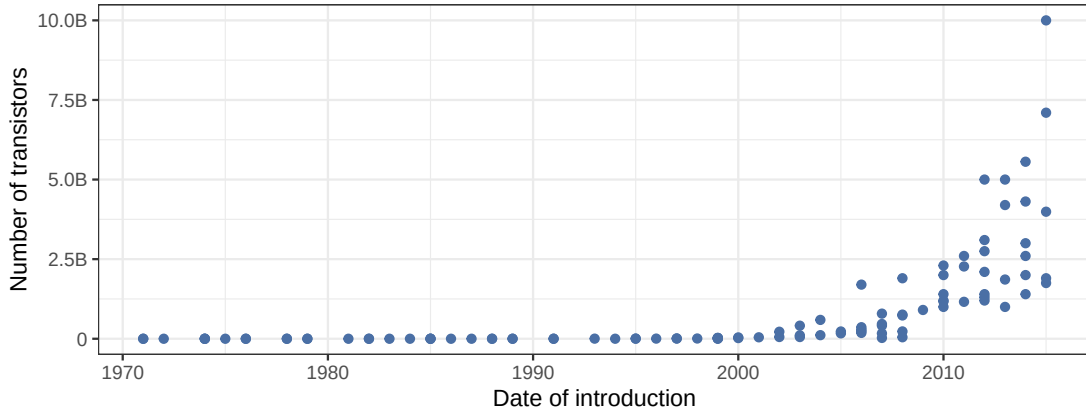
The **log** transformation is frequently useful in regression, because many nonlinear relationships are naturally exponential.

- $\log_b x = y$ when $b^y = x$
- For example, $\log_{10} 1000 = 3$, $\log_{10} 100 = 2$, and $\log_{10} 10 = 1$
- The natural log is $\log_e$, where $e \approx 2.72$ — when we say “log” we will usually mean “natural log” (although for our purposes the base doesn’t matter)



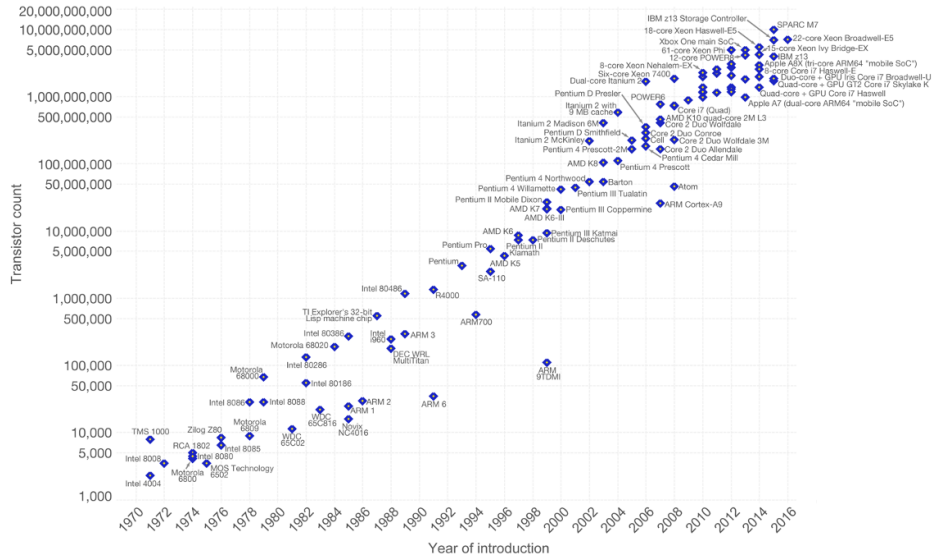
- Moore's Law was a prediction made by Gordon Moore in 1965 (!) that the number of transistors on computer chips would double every 2 years
- This implies **exponential** growth, so a linear model won't fit well (and neither will any polynomial)

Transistor counts over time



Our World
in Data

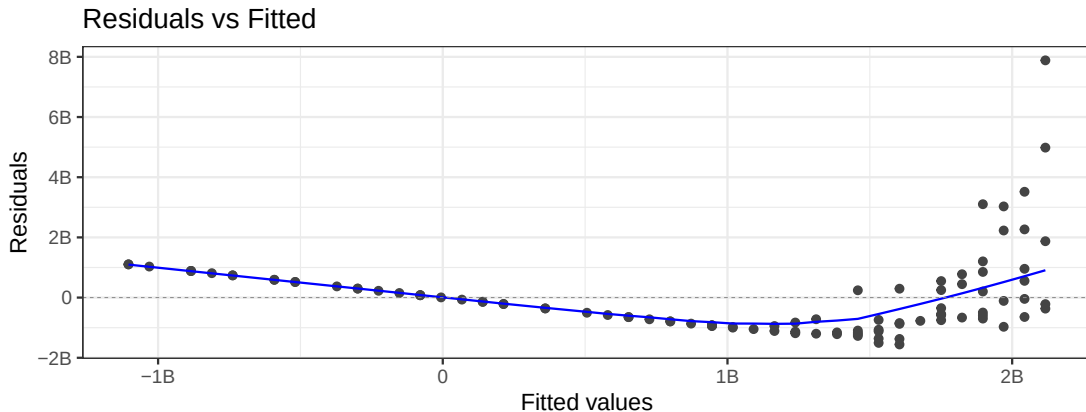
This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are strongly linked to Moore's law.



The data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find more visualizations and research on this topic.

Licensed under [CC-BY-SA](#) by the author Max Roser.

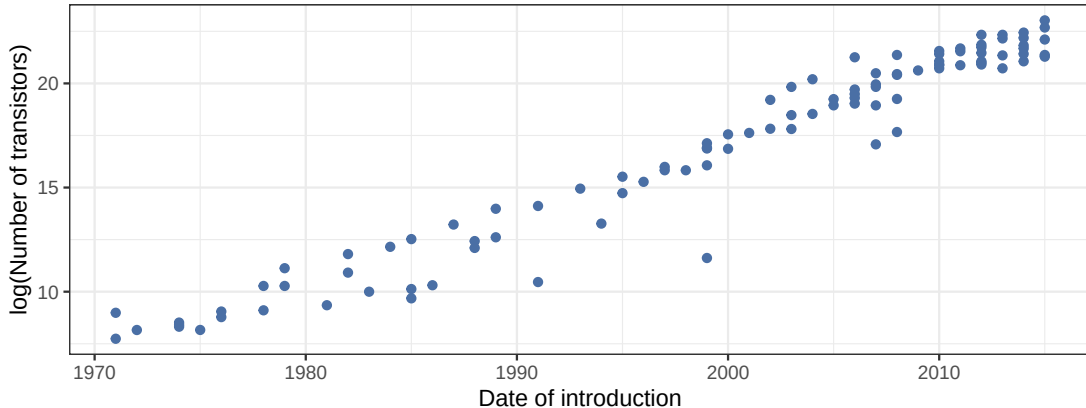
A linear model is a spectacular fail



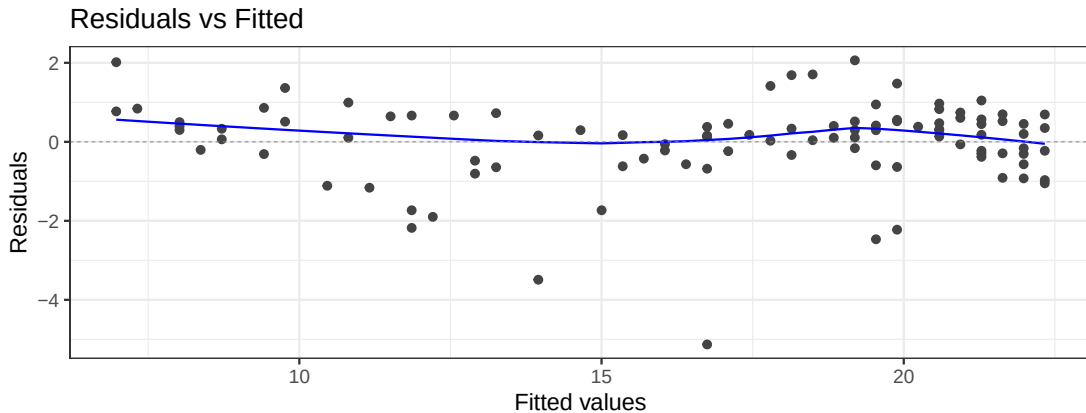


- If $Y = ae^{bX}$, then $\log(Y) = \underbrace{\log(a)}_{\beta_0} + \underbrace{b}_{\beta_1} X$
- In other words, $\log(Y)$ is a linear function of X
- So to model Y as an exponential function of X , predict $\log(Y)$ as a linear function of X

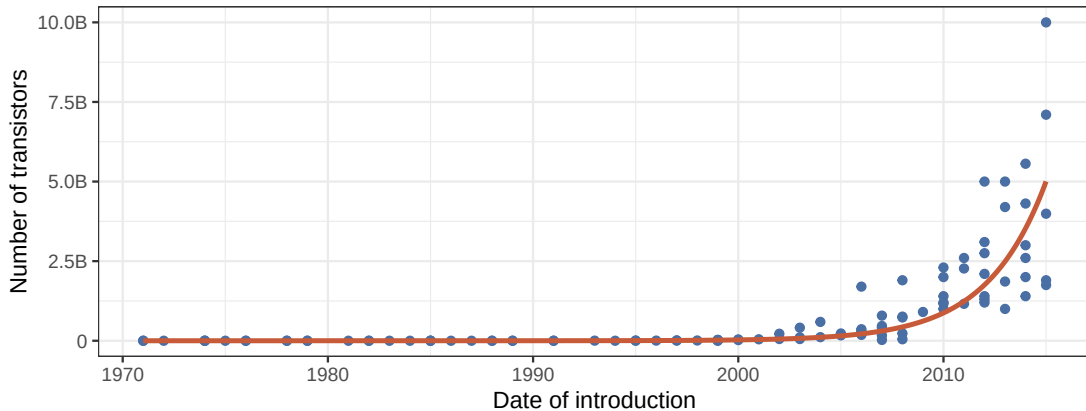
$\log(\text{Transistors})$ is linear in year



A log-linear model satisfies the linearity assumption



Interpreting the log-linear model





Our model is

$$\log(\widehat{\text{Transistors}}) = -681.21 + 0.35 \cdot \text{Year}$$



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- **More useful:** Every year, the predicted number of transistors goes up by 35%



Our model is

$$\log(\widehat{\text{Transistors}}) = -681.21 + 0.35 \cdot \text{Year}$$

Two interpretations of the slope coefficient:

- Every year, the predicted log of transistors goes up by 0.35
- **More useful:** Every year, the predicted number of transistors goes up by 35%
- So the predicted doubling time is $\log(2)/0.3492 = 1.99$ years — we just recovered Moore's “every 2 years” from the data

Model	Equation	Interpretation
Linear	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$	1 unit increase in $X \rightarrow \hat{\beta}_1$ unit increase in $\hat{Y}$
Log-linear	$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 X$	1 unit increase in $X \rightarrow \approx (100\hat{\beta}_1)\%$ increase in $\hat{Y}$
Linear-log	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$	1% increase in $X \rightarrow \approx 0.01\hat{\beta}_1$ unit increase in $\hat{Y}$
Log-log	$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$	1% increase in $X \rightarrow \approx \hat{\beta}_1\%$ increase in $\hat{Y}$

Model	Example	Interpretation
Linear	$\hat{Y} = 2 + 0.3X$	1 unit increase in $X \rightarrow 0.3$ unit increase in $\hat{Y}$
Log-linear	$\log(\hat{Y}) = 2 + 0.3X$	1 unit increase in $X \rightarrow \approx 30\%$ increase in $\hat{Y}$
Linear-log	$\hat{Y} = 2 + 0.3 \log(X)$	1% increase in $X \rightarrow \approx 0.003$ unit increase in $\hat{Y}$
Log-log	$\log(\hat{Y}) = 2 + 0.3 \log(X)$	1% increase in $X \rightarrow \approx 0.3\%$ increase in $\hat{Y}$



- You can transform X , Y , or both
- Anytime you need to “squash” one of the variables (logs make huge numbers not so big!), try transforming it with a log
- In this case, Transistors is skewed right so it is a good candidate for log
- You may need to do a little bit of trial and error to see what works best
- Other transformations are possible!

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- The beer data set contains sales data for cases of beer sold at a supermarket chain
- `PRICE18PK` is the price of 18-packs, and `CASES18PK` is the number of 18-packs sold
- There are also variables for 12-packs and 30-packs

- Price elasticity is the ratio of how much demand (quantity sold) will increase when the price increases:

$$\begin{aligned}\text{elasticity} &= \frac{\% \text{ change in demand}}{\% \text{ change in price}} \\ &= \% \text{ increase in demand when price increases by } 1\%\end{aligned}$$

- Price elasticity is the ratio of how much demand (quantity sold) will increase when the price increases:

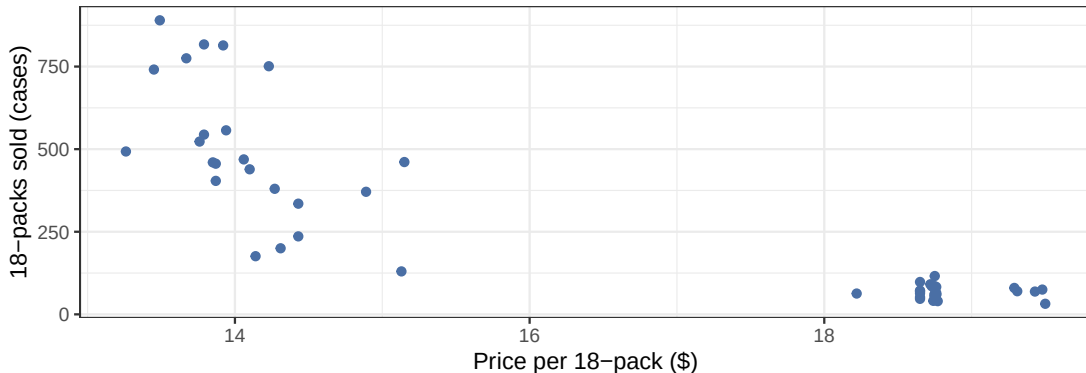
$$\begin{aligned}\text{elasticity} &= \frac{\% \text{ change in demand}}{\% \text{ change in price}} \\ &= \% \text{ increase in demand when price increases by } 1\%\end{aligned}$$

- This is exactly our interpretation of slope coefficients in a log-log model!

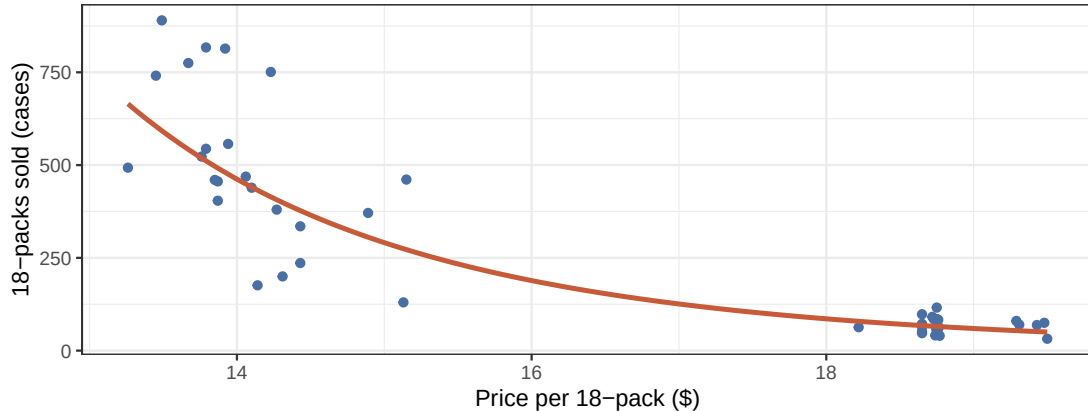
Will a linear model work to predict quantity from price?



This is essentially a demand curve:



Fitting a log-log model



```
log.log.model <- lm(log(CASES18PK) ~ log(PRICE18PK),  
                    data=beer)  
summary(log.log.model)
```

```
Call:  
lm(formula = log(CASES18PK) ~ log(PRICE18PK), data = beer)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-0.8986	-0.2043	-0.0013	0.2665	0.5948

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	23.831	0.946	25.2	<2e-16 ***
log(PRICE18PK)	-6.705	0.337	-19.9	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.36 on 50 degrees of freedom
```

```
Multiple R-squared:  0.888, Adjusted R-squared:  0.886
```

```
F-statistic: 397 on 1 and 50 DF,  p-value: <2e-16
```



Our fitted model is

$$\log(\widehat{\text{Cases}}) = 23.83 - 6.71 \cdot \log(\text{Price})$$



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- Demand is **wildly elastic**: shoppers substitute to other pack sizes and other brands the moment the price moves



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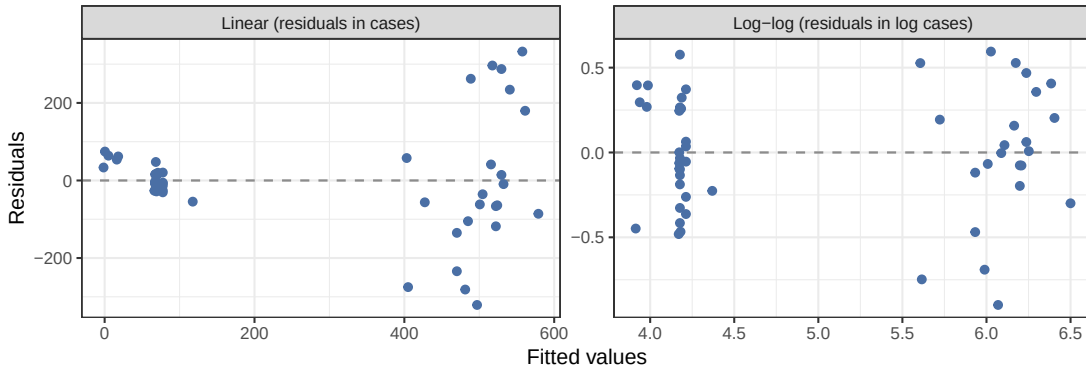
$$\log(\widehat{\text{Cases}}) = 23.83 - 6.71 \cdot \log(\text{Price})$$

- A 1% increase in price is predicted to drop 18-pack sales by about 6.7%
- Demand is **wildly elastic**: shoppers substitute to other pack sizes and other brands the moment the price moves
- Compare the 12-packs, where the same model gives an elasticity of only -4.0 — pack size changes the story

Log transformations can also fix heteroscedasticity!



Residuals for the linear model (left) vs the log-log model (right):





- **Polynomials** bend the line when the relationship curves but has no natural multiplicative structure — and buy very little past the first bend



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- The transformation changes the **interpretation**, not just the fit: percentages and elasticities instead of units



- **Polynomials** bend the line when the relationship curves but has no natural multiplicative structure — and buy very little past the first bend
- **Logs** are the right tool when growth is exponential or effects are multiplicative, and they often fix heteroscedasticity as a bonus
- The transformation changes the **interpretation**, not just the fit: percentages and elasticities instead of units
- Always ask what the model does **outside** the range of your data